Cold Chain Logistics Dashboard

User Guide: Shipment Monitoring and Quality Assessment

a. Shipment Summary

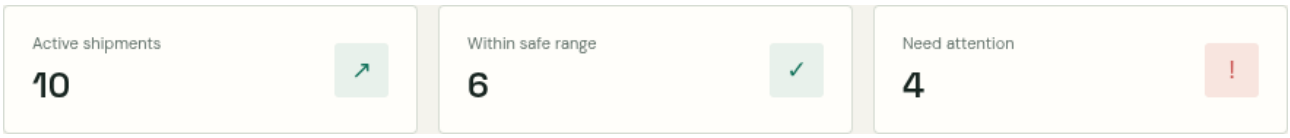


Figure 1. Shipment summary counters.

Function: Summarizes the shipment count, shipments within their safe temperature ranges, and shipments needing attention.

Steps:

1. Read **Active shipments** to see the number of records on the dashboard.
2. Compare **Within safe range** with **Need attention**.
3. For a shipment needing attention, locate its ID in the activity table and review its temperature and history.

b. Creating a Shipment

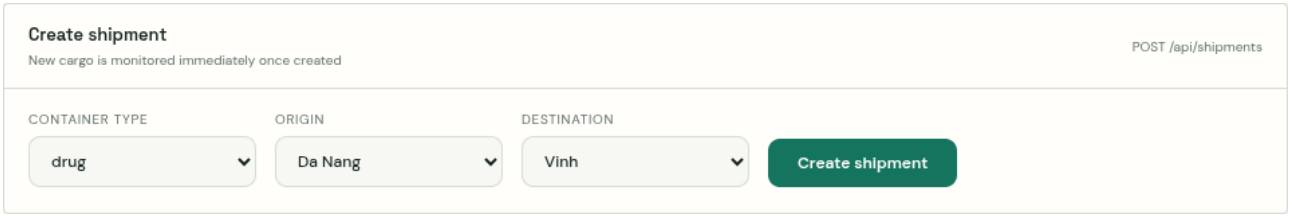


Figure 2. Create shipment form.

Function: Adds a new shipment to the monitoring dashboard.

Steps:

1. Select the container type, origin, and destination.
2. Select **Create shipment** once and wait for the result message below the form.
3. After a successful response, find the new shipment in the activity table.

c. Monitoring Shipment Location

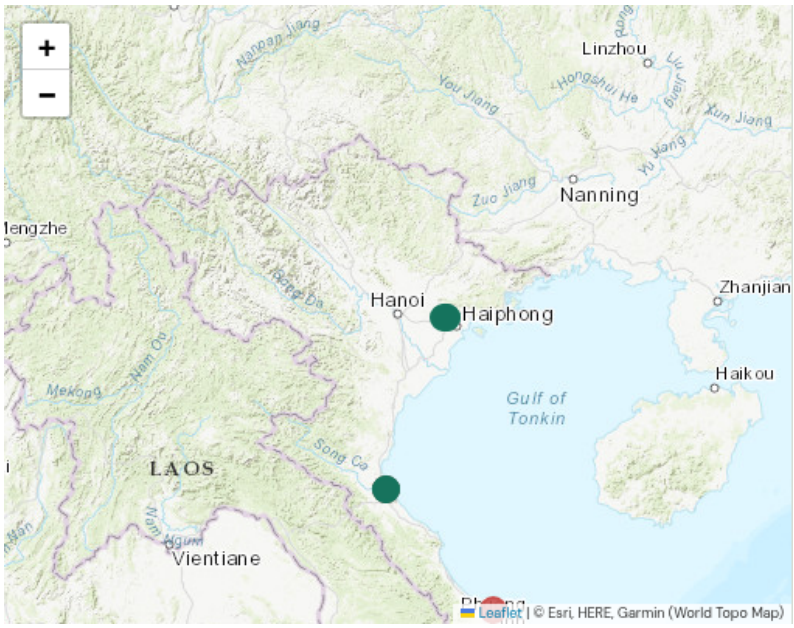


Figure 3. Live shipment map and route markers.

Function: Shows current shipment positions, selected routes, and destinations on the map.

Steps:

1. Select a shipment row in the activity table.
2. Review the colored shipment marker and highlighted route.
3. Use the map controls to zoom in or out and inspect the route.

d. Reviewing Temperature History

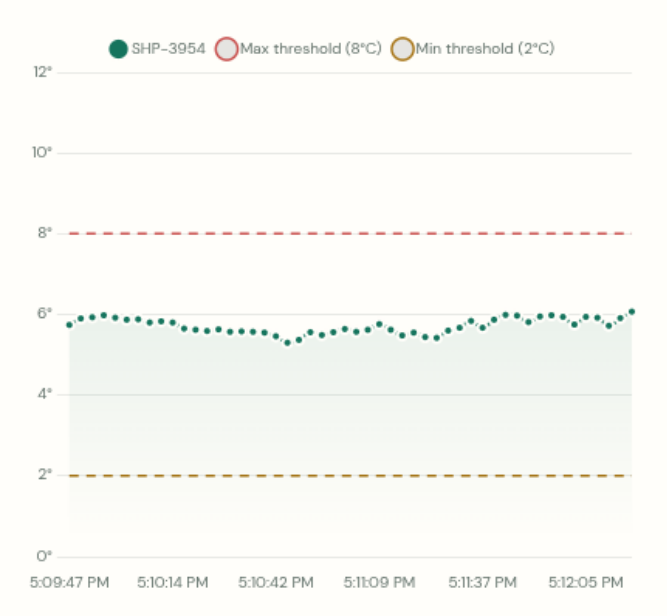


Figure 4. Temperature history chart for the selected shipment.

Function: Plots sensor readings against the selected container’s minimum and maximum temperature limits.

Steps:

1. Select the shipment to review from the activity table.
2. Compare the measured temperature line with the dashed limit lines.
3. Use the chart legend and time labels to interpret the readings.

e. Searching and Filtering Shipments

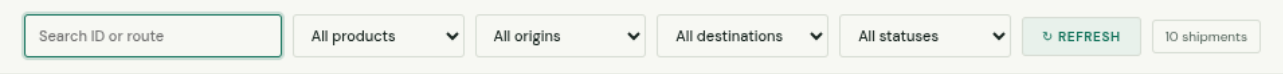


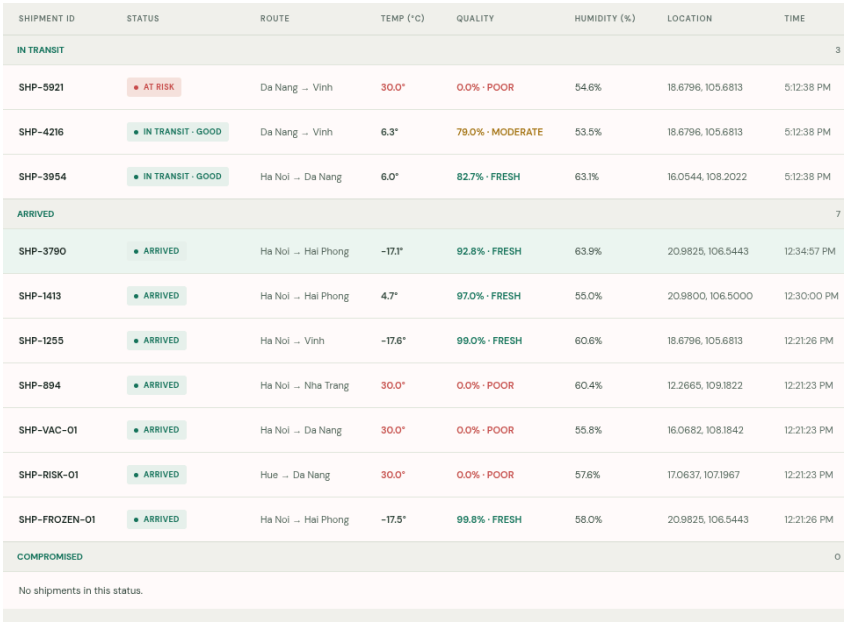
Figure 5. Shipment search and filter controls.

Function: Filters the shipment table by shipment ID or route, product, origin, destination, and status.

Steps:

1. Enter a shipment ID or route name in the search field.
2. Select a product, origin, destination, or status. Multiple filters can be combined.
3. Review the result count. Select the “All” option to clear a filter.
4. Select **Refresh** to reload shipment and telemetry data.

f. Assessing Shipment Quality



SHIPMENT ID	STATUS	ROUTE	TEMP (°C)	QUALITY	HUMIDITY (%)	LOCATION	TIME
IN TRANSIT							3
SHP-5921	AT RISK	Da Nang → Vinh	30.0°	0.0% · POOR	54.6%	18.6796, 105.6813	5:12:38 PM
SHP-4216	IN TRANSIT · GOOD	Da Nang → Vinh	6.3°	79.0% · MODERATE	53.5%	18.6796, 105.6813	5:12:38 PM
SHP-3954	IN TRANSIT · GOOD	Ha Noi → Da Nang	6.0°	82.7% · FRESH	63.1%	16.0544, 108.2022	5:12:38 PM
ARRIVED							7
SHP-3790	ARRIVED	Ha Noi → Hai Phong	-17.1°	92.8% · FRESH	63.9%	20.9825, 106.5443	12:34:57 PM
SHP-1413	ARRIVED	Ha Noi → Hai Phong	4.7°	97.0% · FRESH	55.0%	20.9800, 106.5000	12:30:00 PM
SHP-1255	ARRIVED	Ha Noi → Vinh	-17.6°	99.0% · FRESH	60.6%	18.6796, 105.6813	12:21:26 PM
SHP-894	ARRIVED	Ha Noi → Nha Trang	30.0°	0.0% · POOR	60.4%	12.2665, 109.1822	12:21:23 PM
SHP-VAC-01	ARRIVED	Ha Noi → Da Nang	30.0°	0.0% · POOR	55.8%	16.0682, 108.1842	12:21:23 PM
SHP-RISK-01	ARRIVED	Hue → Da Nang	30.0°	0.0% · POOR	57.6%	17.0637, 107.1967	12:21:23 PM
SHP-FROZEN-01	ARRIVED	Ha Noi → Hai Phong	-17.5°	99.8% · FRESH	58.0%	20.9825, 106.5443	12:21:26 PM
COMPROMISED							0
No shipments in this status.							

Figure 6. Quality scores in the shipment activity table.

Function: Estimates quality from the latest temperature's deviation from the container's target midpoint.

Steps:

1. Find the shipment's latest temperature and its Quality score.
2. Interpret the score: **FRESH** is 80% or higher; **MODERATE** is 50% to below 80%; **POOR** is below 50%.
3. Compare the result with the container temperature limits and any active alert.

g. Reviewing Completed Shipment History

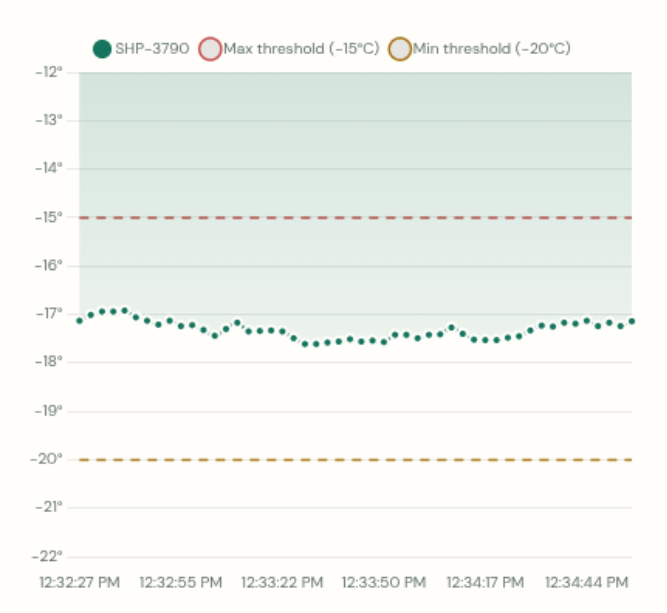


Figure 7. Historical readings for completed shipment SHP-3790.

Function: Keeps completed shipments available for review while stopping new live updates for those records.

Steps:

1. Select **Completed** in the status filter.
2. Select the shipment row to view its route and historical temperature chart.
3. Confirm that the chart subtitle says **Historical readings**.
4. The dashboard shows up to 50 newest completed shipments and up to 50 readings per shipment. Older database records are not deleted.

h. Connection and Troubleshooting

Function: Helps restore dashboard data and live monitoring when the backend connection is unavailable.

Steps:

1. Confirm the header says “Connected to backend”.
2. If connection fails, confirm that the backend and PostgreSQL are running.
3. Use **Refresh** after the connection is restored.
4. If map tiles are missing while shipment data is present, check the Internet connection.

Dashboard URL: http://127.0.0.1:5500/?apiPort=8081&wsPort=8080